Module 01: Systems — Neural Systems as Dynamical Systems

> **Course**: Active Inference 101 | **Unit**: Computational Neuroscience | **Audience**: First-semester undergraduates

Learning Objectives

1. Describe the brain as a **dynamical system** — a system whose state changes over time according to rules.
2. Explain how **neural populations** form self-organizing patterns through excitation and inhibition.
3. Introduce the concept of **attractors** — stable patterns that neural activity settles into.

Introduction

The brain is the most complex system we know. It contains roughly 86 billion neurons, each connected to thousands of others, creating a network of staggering complexity. How do we even begin to understand it? One powerful approach is to view the brain as a **dynamical system** — a system whose state evolves over time following certain rules. This module introduces the computational neuroscience perspective on systems.

Key Concepts

1. Neurons as Information Processors

A single neuron performs a basic computation:

- **Inputs**: Receives signals from other neurons through **dendrites**
- **Integration**: Sums up excitatory (+) and inhibitory (−) inputs at the **cell body**
- **Output**: If the sum exceeds a threshold, the neuron **fires** — sending an electrical pulse (action potential) down its **axon** to other neurons

But individual neurons aren't very interesting. The magic happens in **populations** — large groups of neurons working together.

2. Neural Populations and Oscillations

Groups of neurons naturally produce **oscillations** — rhythmic patterns of firing:

- **Alpha waves (~10 Hz)**: Present when you're relaxed with eyes closed
- **Beta waves (~20 Hz)**: Active during focused thinking and motor planning
- **Gamma waves (~40 Hz)**: Associated with attention, binding, and consciousness
- **Theta waves (~5 Hz)**: Prominent in the hippocampus during memory formation

These oscillations aren't random — they're self-organized patterns that emerge from the interactions between excitatory and inhibitory neurons. They're the brain's way of coordinating activity across regions.

3. The Brain as a Dynamical System

A **dynamical system** is anything whose state changes over time according to rules. The brain is one:

- **State**: The pattern of neural activity at any moment (which neurons are firing, how fast)
- **Rules**: The connections between neurons (synaptic weights), the properties of ion channels, neurotransmitter concentrations
- **Trajectory**: How neural activity evolves from one moment to the next

This perspective lets us use mathematical tools from physics and engineering to understand the brain.

4. Attractors — Where Neural Activity Settles

An **attractor** is a pattern of activity that the system tends to settle into, like a ball rolling to the bottom of a valley:

- **Fixed-point attractor**: The system converges to a single stable state (e.g., resting neural activity)
- **Limit-cycle attractor**: The system settles into a repeating loop (e.g., neural oscillations)

- **Strange attractor**: The system follows a complex, non-repeating but bounded trajectory (e.g., chaotic neural dynamics)

Memory recall can be understood as the brain's activity being attracted toward a stored pattern. When you see part of a friend's face, your brain's dynamics are "attracted" toward the full representation.

5. From Neural Dynamics to the Markov Blanket

The dynamical systems perspective connects to the Markov Blanket from Module 01 (Cognitive Science). A neuron's Markov Blanket consists of:

- **Sensory states**: Inputs from other neurons (dendrites)
- **Active states**: Outputs to other neurons (axon terminals)
- **Internal states**: The neuron's membrane potential, ion channel states

At a larger scale, a brain region's Markov Blanket separates it from other regions, with particular input and output pathways. This hierarchical structure is central to how the brain implements Active Inference.

Summary

The brain is a dynamical system composed of neural populations that self-organize into oscillatory patterns. Attractor dynamics explain how neural activity stabilizes into meaningful patterns (memories, percepts, motor plans). The Markov Blanket concept applies at every scale — from individual neurons to brain regions to the whole organism.

Further Reading

- Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature Neuroscience*, 20(3), 340-352.
- Deco, G., Jirsa, V. K., & McIntosh, A. R. (2011). Emerging concepts for the dynamical organization of resting-state activity in the brain. *Nature Reviews Neuroscience*, 12(1), 43-56.
- Strogatz, S. H. (2015). *Nonlinear Dynamics and Chaos* (2nd ed.). Westview Press.